



A²LC: Active and Automated Label Correction for Semantic Segmentation

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Introduction

- Task: **Label acquisition for semantic segmentation**
 - Challenge: High cost and labor-intensive nature of pixel-wise annotation
 - Promising solution: Active Label Correction with foundation models (ALC)
- Two key challenges of **ALC**
 - Redundant queries for similar patterns
 - Class imbalance challenge
- Main contributions of **A²LC**
 - Label Correction Module (**LCM**) automatically corrects noisy labels exhibiting similar features with human-corrected masks.
 - Adaptively Balanced Confidence in label (**ABC**) increases the sampling frequency of tail classes.

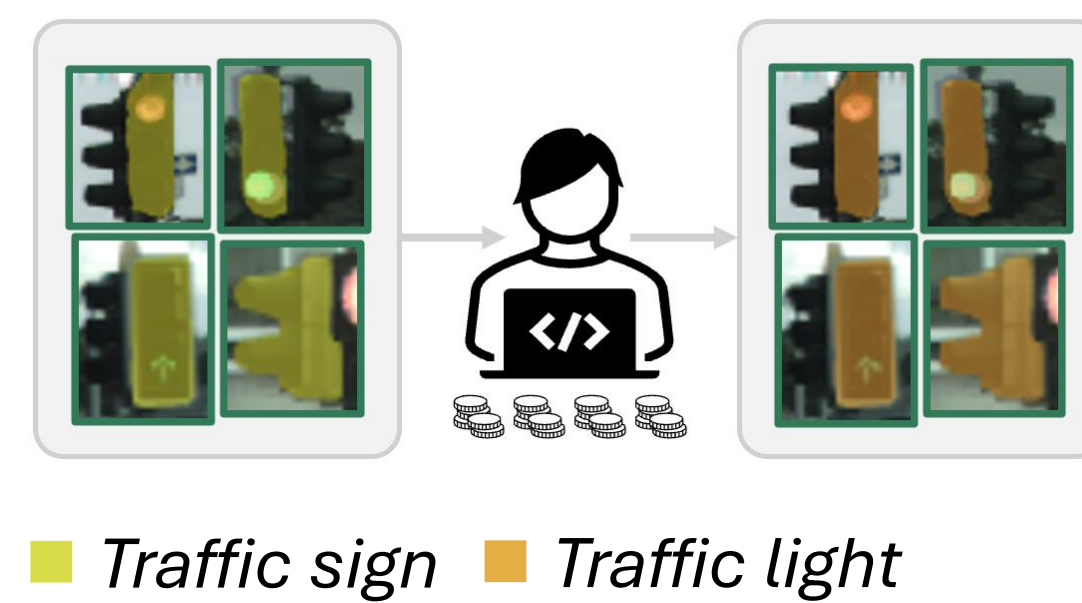
Preliminaries: ALC Framework

Overview of ALC framework

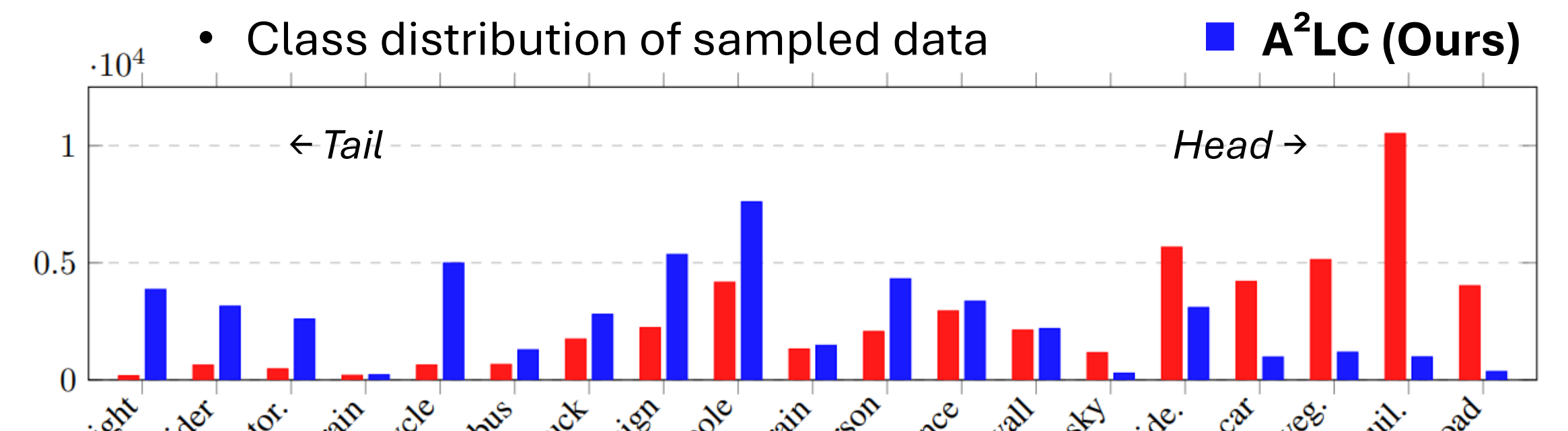
- Initial pseudo-labels are generated by the foundation model.
- In each round, top- B informative pixels are sampled by acquisition function using deep learning model and queried to annotator.
- After manual corrections, both the pseudo-labels and model are updated, completing a single round of the correction cycle.

Two key challenges of ALC

- Redundant queries lead to inefficient resource utilization.



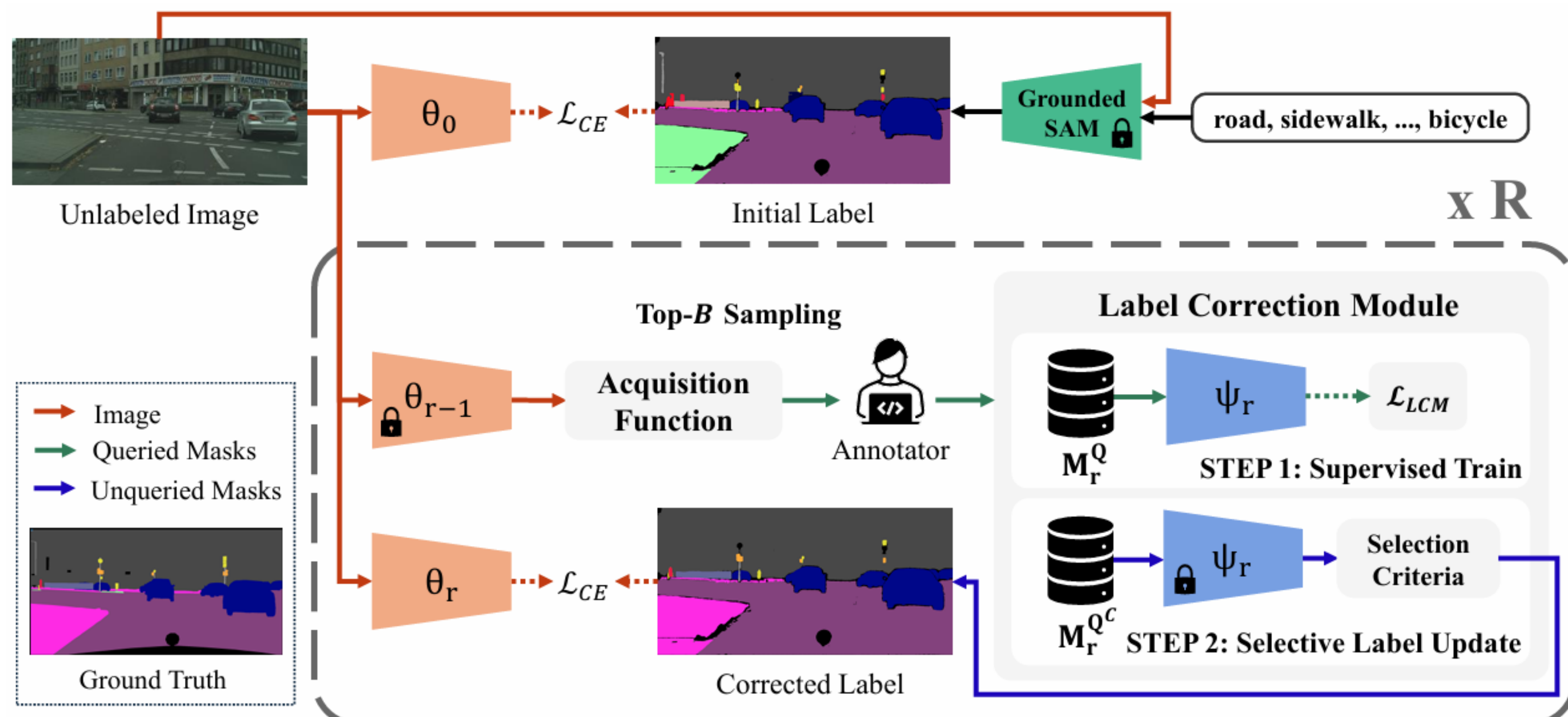
- Sampling bias toward head classes results in suboptimal corrections.



A²LC Framework

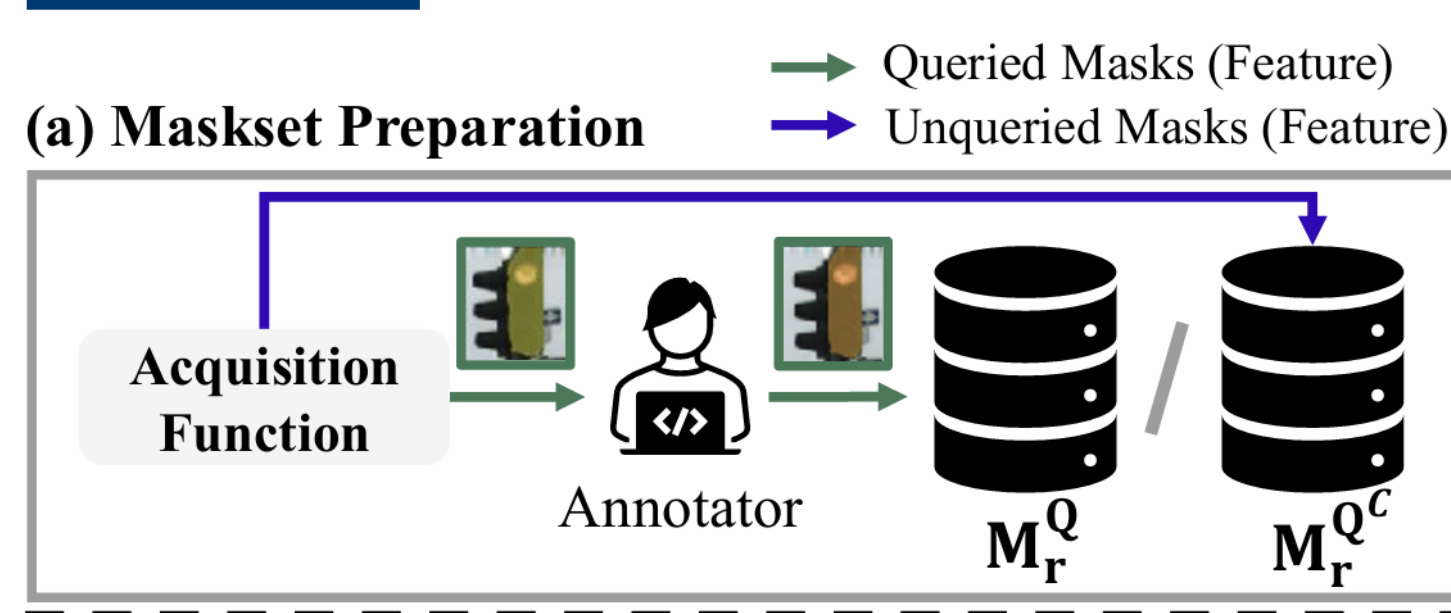
Overview of A²LC framework

- A²LC framework is a semi-automated label correction framework for semantic segmentation built upon cascading stages of manual and automatic correction.



Main contributions of A²LC

1. LCM



- LCM performs automatic correction by propagating human-provided labels beyond the queried samples.

- LCM correction is applied only to the masks that satisfy all selection criteria.

- Selection criteria** are designed to identify predictions with high reliability, so that automatic label updates are applied exclusively to the subset of confidently inferred labels.

$$\mathcal{J}^m = \mathcal{J}_1^m \wedge \mathcal{J}_2^m \wedge \mathcal{J}_3^m$$

$$\mathcal{J}_1^m = \mathbb{I} \left(\max_{c \in \mathcal{C}} \psi_r(c; m) \geq \tau \right)$$

$$\mathcal{J}_2^m = \mathbb{I} \left(\hat{y}_{\psi_r}(m) \notin \{c \mid \text{rank}(c) \geq (1 - \alpha) \cdot |\mathcal{C}|, c \in \mathcal{C}\} \right)$$

$$\mathcal{J}_3^m = \mathbb{I} \left(\hat{y}(m) \neq \arg \max_{c \in \mathcal{C}} \text{rank}(c) \right)$$

2. ABC

- ABC guides both correction stages toward improved class balance by incorporating pixel-wise adaptive class weight into the acquisition function.

$$a_{ABC}(m; \theta) := \sum_{x \in m} \frac{f_{\theta}(x) \cdot f_{\theta}(m')}{\|f_{\theta}(x)\| \|f_{\theta}(m')\|} \cdot a_{ABC}(x; \theta) \quad a_{ABC}(x; \theta) := w(x) \cdot a_{CIL}(x; \theta)$$

- Adaptive class weight** is composed of two components, class rarity score and dataset imbalance score.

- Class rarity score** prioritizes pixels belonging to tail classes during sampling.
- Dataset imbalance score** adaptively emphasizes the class rarity score according to the pseudo-label statistics at each round.

$$w(x) := \hat{w}(x)^{\text{KL}^3(\mathbb{P}_{\text{dist}} \| \mathbb{U}_{\text{dist}})} \quad \hat{w}(x) := \frac{\min_{c \in \mathcal{C}} |\{x' \in \mathcal{M} : \hat{y}(x') = c\}|}{|\{x' \in \mathcal{M} : \hat{y}(x') = \hat{y}(x)\}|} \quad \text{KL}(\mathbb{P}_{\text{dist}} \| \mathbb{U}_{\text{dist}}) = \sum_{c \in \mathcal{C}} \mathbb{P}(c) \log \frac{\mathbb{P}(c)}{\mathbb{U}(c)}$$

Experimental Results

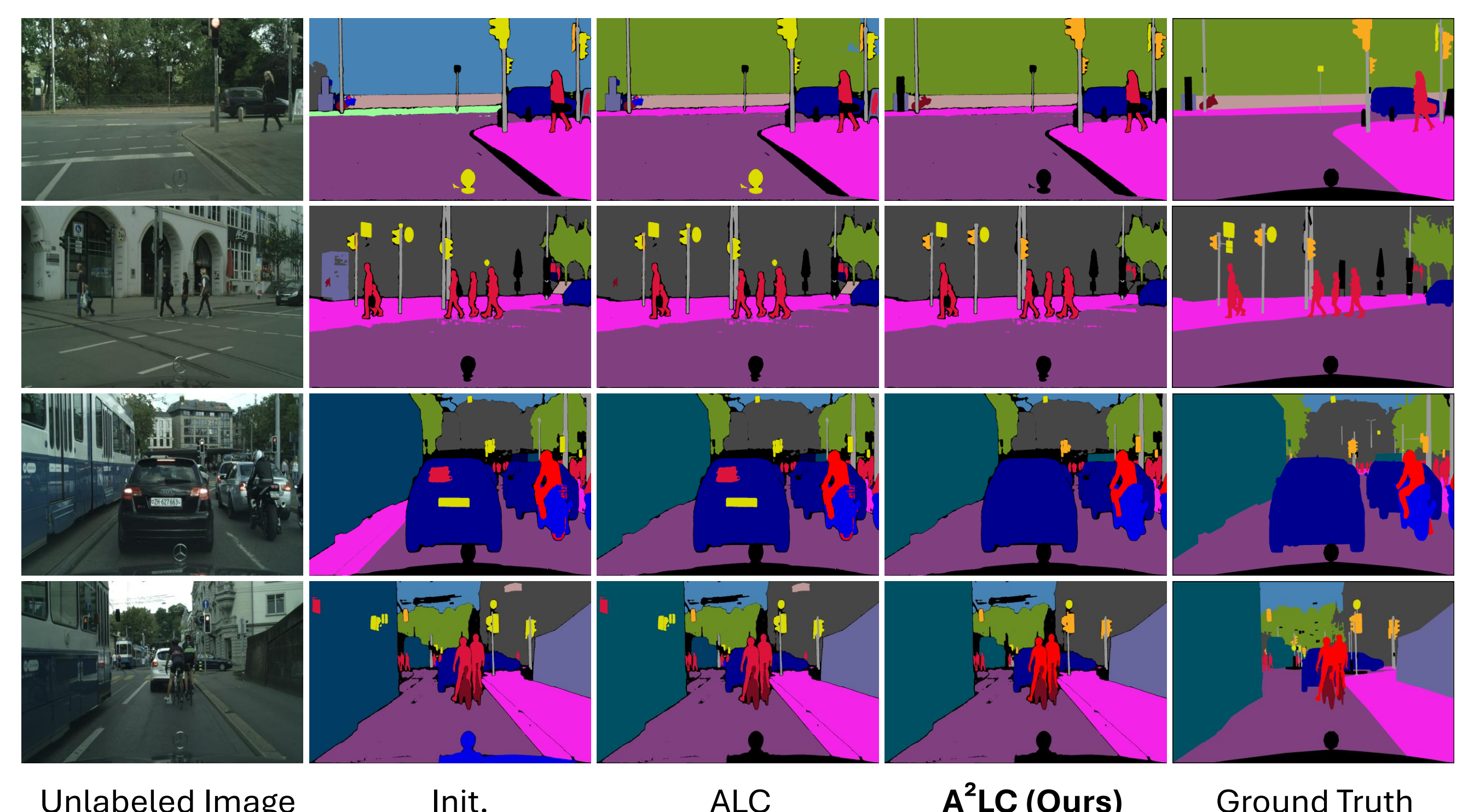
Quantitative results

- High efficiency:** Requires only 20% (Cityscapes) and 60% (PASCAL) of the budget.
- Strong effectiveness:** Achieves +27.23% (Cityscapes) and +14.30% (PASCAL) performance gains under the same budget.

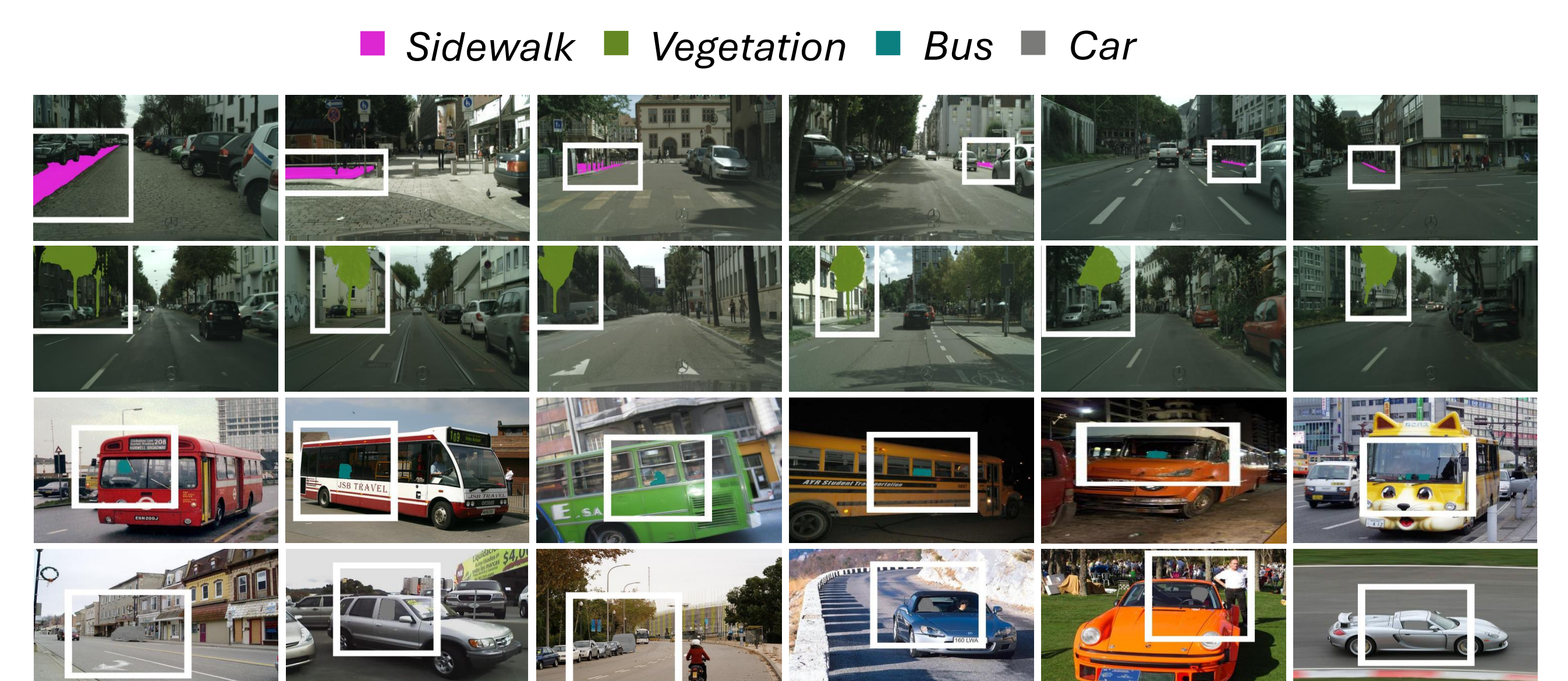
Dataset	mIoU (%)	Methods	Init.	1R	2R	3R	4R	5R
Cityscapes	Data	ALC	50.68 $\pm$ 0.00	62.28 $\pm$ 1.25	65.36 $\pm$ 0.56	66.19 $\pm$ 0.60	66.87 $\pm$ 0.29	67.01 $\pm$ 0.30
		A ² LC (Ours)	50.68 $\pm$ 0.00	70.01 $\pm$ 1.05	75.60 $\pm$ 0.58	80.84 $\pm$ 0.53	82.69 $\pm$ 0.92	85.26 $\pm$ 0.55
	Model	ALC	51.55 $\pm$ 0.71	56.61 $\pm$ 0.42	58.27 $\pm$ 0.63	58.58 $\pm$ 0.08	58.52 $\pm$ 0.26	58.59 $\pm$ 0.05
		A ² LC (Ours)	51.55 $\pm$ 0.71	60.83 $\pm$ 0.66	63.89 $\pm$ 0.44	67.50 $\pm$ 0.31	68.87 $\pm$ 0.78	70.51 $\pm$ 0.32
PASCAL	Data	ALC	58.63 $\pm$ 0.00	68.19 $\pm$ 0.34	72.72 $\pm$ 0.16	74.84 $\pm$ 0.29	76.41 $\pm$ 0.70	77.06 $\pm$ 0.73
		A ² LC (Ours)	58.63 $\pm$ 0.00	67.49 $\pm$ 2.03	74.88 $\pm$ 1.45	80.88 $\pm$ 0.83	84.81 $\pm$ 0.48	88.08 $\pm$ 0.44
	Model	ALC	56.94 $\pm$ 0.44	62.11 $\pm$ 0.61	64.12 $\pm$ 0.31	64.15 $\pm$ 0.68	65.00 $\pm$ 0.33	65.48 $\pm$ 0.83
		A ² LC (Ours)	56.94 $\pm$ 0.44	60.87 $\pm$ 2.93	64.08 $\pm$ 2.43	66.45 $\pm$ 0.95	67.76 $\pm$ 0.06	68.42 $\pm$ 0.87

Qualitative results

- Constructed pseudo-labels



- Visualization of LCM-corrected masks



Ablation study

Methods			Data mIoU (%)					Model mIoU (%)				
LCM	ABC	Mask	1R	2R	3R	4R	5R	1R	2R	3R	4R	5R
✓			62.60	65.20	66.03	66.85	66.86	56.86	58.93	58.63	58.59	58.59
	✓		65.66	71.53	76.00	78.38	80.59	59.02	61.39	64.19	65.58	66.94
		✓	67.31	70.06	74.49	78.32	81.58	59.09	60.86	62.05	65.27	67.75
✓	✓		70.62	77.29	80.39	82.80	84.15	60.32	65.09	67.36	68.71	70.11
✓	✓	✓	71.04	76.08	81.13	83.53	85.57	61.06	64.39	67.83	69.63	70.88